



Citrate,
Heparin and no
anticoagulant
use

B. BRAUN OMNI

Troubleshooting guide

OUR VISION

We
PROTECT
and
IMPROVE
the
HEALTH
of people
around the world.

CONTENT

Power on and off

Set up

Priming

Entering parameters

Prior to start of therapy

Starting therapy

Common troubleshooting during therapy

Temporary disconnection and recirculation

Change of kit

Ending therapy

This guide is linked to the OMNI 1.75 software and IFU. This guide does not replace the OMNI IFU SW 1.75 and is recommendation only. It should be used in conjunction with appropriate training as an additional help guide.

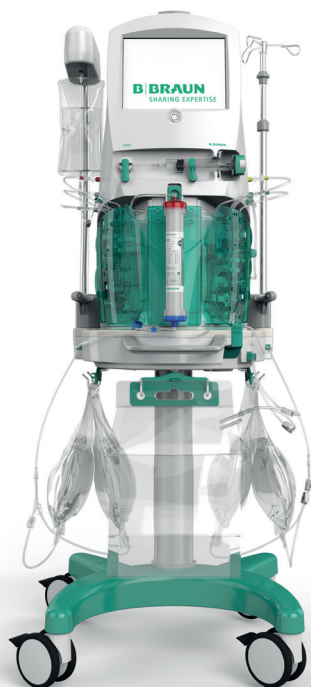
Any issues with patient electrolytes, protocol or clinical issues please refer to your consultant/pdn/senior nursing staff.

The OMNI technical helpline is for reporting technical problems with the machine.

Technical helpline: 0114 225 9204
(available Mon – Fri, 8:30am – 4:30pm)

Out of hours: 0114 225 9299

ALWAYS ensure the OMNI
is plugged in,
even when not in use





POWER ON AND OFF



**ALWAYS
KEEP**



**PLUGGED
IN**



ALWAYS keep the OMNI plugged in, even when not in use. If the OMNI does not switch on, it is likely to have run out of charge. Plug in and allow to charge before retrying.



To turn ON

Short press



To turn OFF

Press and hold



To switch the OMNI on, the green power button should only be pressed once for a short time. Do not press and hold as this will power the machine on and then off. To switch off the OMNI, press and hold the green button.



**FULL
CHARGE**

=

30 minutes

The OMNI has a battery capacity of 30 minutes. Therapy stops when the machine is unplugged.



SET UP



Ensure to click both top and bottom at both fluid and blood sides when inserting set.

PRESSURE LINES



The 'Connect Pressure Lines' action does not have an hourglass or get a tick when the pressure lines are attached. There will be a tick once the machine has loaded the set, always follow each instruction in turn.

LOADING



Let the machine load itself, never try and load the lines manually and never try to unload the set manually. Pulling the lines out of the pump manually can result in breakages in the pumps.

SYRINGE LINES



ALWAYS choose the correct syringe line when inserting the syringe when using OMNISET L or PRO. Calcium Chloride is ALWAYS attached to the line that says Calcium line, which is always used with CITRATE anticoagulation. The HEPARIN syringe is always attached to the long thin syringe line that has no writing on it. It is imperative to ALWAYS choose the right line.



If using CVVHD then always put one bag of saline (either 500 ml or 1000 ml) onto the purple line/substitution side, this is for priming of the purple line only and stays there for the entirety of the therapy. In CVVH and CVVHDF there is a substitution bag on this scale for post dilution and for these therapies we can connect two substitution bags to this scale.



Different fluid bags are used for either CVVHD or CVVHDF with citrate anticoagulation. Ensure to always follow the protocol and prescription carefully when setting up and changing fluid bags on the OMNI.



SET UP



Frangible pins: ensure to make sure all pins on fluid bags are broken correctly. If there is a flow problem/pump failure/solution pressure/empty priming bag related alarm, this COULD be caused by unbroken or partially broken pins, please always double check that they are fully broken.



All scales are separate and do not need to be balanced. It is worthwhile putting two fluid and effluent bags on at the start, to reduce workload.



PRIMING

Prior to priming: ensure all unused clamps are closed and all used are open before pressing Automatic Priming. When using heparin or no anticoagulant and using the OMNISET L or PRO, ensure to close the clamps on syringe lines not being used AND the white clamp on the calcium line at the top of the blue venous/return line. This line does not get primed in heparin, therefore it is imperative that it is closed.

Press automatic prime and leave to prime.

'Arterial Line Blocked' may appear and if so, push the spike into the priming bag further attached to the red access line and press reset alarm. This should resolve this alarm, if the alarm appears again press reset again and it should continue to prime.

If there are any flow problem alarms or any alarms relating to fluids, then double check the clamps and pins.



ENTERING PARAMETERS



This can be done during priming, or prior to the patient being put onto the OMNI. Be careful not to move the machine during priming to avoid any errors.



It is important to ensure the Calcium Chloride and Citrate used adhere to the concentrations set on the OMNI to ensure the volumes being delivered by the machine are correct.



When using citrate we must not change the blood flow unless stated on the protocol or informed so by a Consultant. If we change our blood flow, we are altering the amount of citrate we are giving. Consult medical/senior nursing staff regarding queries over the citrate protocol.

THERAPY TIME LIMIT



When entering parameters it is advisable to leave the 'therapy time limit' section as a dash, not entering anything into this box to allow the machine to run for the lifetime of the set. Therapy time limit does not appear on the Confirm Therapy Parameters page.

NET FLUID REMOVAL



Enter the net fluid removal as zero or desired flow prior to start of therapy. Please be careful to enter net fluid removal (fluid to be removed in mls/hr) and the dialysate flow in the correct boxes. Ensure to follow the protocol carefully and check parameters that have been input to avoid any potential errors.



PRIOR TO START OF THERAPY

RINSING

If we need to re-prime the red/access and blue/return lines with saline following initial priming then we do this by rinsing. Attach a new bag of saline to the red access line where the priming bag is and enter the amount needed to rinse.

RECIRCULATION

We can recirculate prior to commencing therapy. To do this attach the access/red line to the three way tap on the waste bag (next to the return/blue line), CLOSE THE WHITE CLAMP to avoid sterilisation chemicals that will be present in the waste line bag from priming being recirculated and then press recirculate. Always leave the machine on during recirculation.



STARTING THERAPY



When enter therapy is pressed please start the blood pump, increase the blood flow to protocol target blood flow at a rate based on the patient's stability.

LOW VENOUS ALARM



If you experience this alarm, please double check your connections as it can happen when the lines become disconnected. However, if the blood lines have not been filled during the patient connection page it may occur whilst the blood is making its way around the set from the patient. This can be due to the saline being returned to the patient at present (from priming) being less viscous than blood and triggering a low venous alarm. If the connection is secure, we can try and raise the height of the bed which can increase the venous pressure and press reset alarm. Once the blood is returning to the patient the problem should be resolved.



Once we have started the blood pump and therapy using the red and blue buttons in the bottom corner, please avoid from pressing these unless required to stop the blood pump or therapy. During therapy if the machine needs to stop them it will and it will similarly restart these itself. For example, during a bag change the therapy will stop and by pressing of the 'bag change/weight reset' button the therapy will then restart itself.



COMMON TROUBLESHOOTING DURING THERAPY

BAG CHANGES

ALWAYS press the 'bags' button when changing any fluid bag. If done without pressing the button, it will trigger an alarm and could cause a 'safety check error' message to show and a possible inability to continue treatment. If this alarm shows and is unable to be reset you may need to make a power failure on the machine (turning it off and then on), this should resolve the alarm. The OMNI will take approximately 3 minutes to switch back on and will return to the last state after restart. Always press the 'Bags' button to ensure this does not occur.

LONG ANTICOAGULATION STOP

This alarm happens when the anticoagulation has been stopped for approximately 5 minutes. The anticoagulation stops when doing a bag or syringe change, always ensure to preempt bag or syringe changes to ensure the set does not clot.

AIR IN CALCIUM SYRINGE

Ensure to use a small syringe and slowly pull the air out, it should only be a small volume required to be removed. Keep the calcium line clamp open when doing this to be able to pull the air out. Do not pull too hard or there will be blood returned in the calcium line from the venous line.



COMMON TROUBLESHOOTING DURING THERAPY



BLOOD LEAK DETECTOR ALARM

1

There is visible blood in the effluent caused by a crack or problem with the filter itself – a new kit is required as it is faulty. Do not return the blood in this case.

2

It can additionally trigger when there is a discolouration of the effluent (for example from high bilirubin levels), if you are satisfied that this is the reason you can press 'temporary override', causing the small box to show in the left hand bottom corner of the screen.

3

Additionally, if there is bright light shining on the blood leak detector (underneath the effluent chamber) this can trigger the alarm, remove the light source and press 'no override'.

4

Finally, if the effluent chamber is empty, this can trigger the alarm. Go to the 'levels' button on the screen and VERY SLOWLY fill the effluent chamber using the up arrow until the fluid is half way only. Do not do this too fast or fill the chamber too full as this can cause a change of pressure within the set and possible subsequent alarms. Once the chamber is filled half way, press 'no override' on the blood leak detector and this should resolve the problem.

If the blood leak detector is put into override, please consider testing the effluent for blood using the small line at the fluid side of the machine to draw out some effluent.



COMMON TROUBLESHOOTING DURING THERAPY

BLOOD IN THE ARTERIAL (RED) PRESSURE LINE

This is due to stops in the blood pump/low blood flow causing a back flow of blood from the arterial/access line. When blood is seen in the arterial pressure line try and resolve this by:

- 1 Pressing 'patient care button'
- 2 Clamping the access/arterial line at patient access
- 3 Removing the red arterial pressure line from the pressure port on the side of the machine only a small amount
- 4 Check to see if the blood has returned to the chamber from the pressure line
- 5 Re-attach the red pressure line to the pressure port on the side of the machine
- 6 Unclamp the access/arterial line at patient access
- 7 Take off patient care mode by pressing the 'patient care' button again

As long as the blood pump is running this will cause the blood to move out of the line and return to the chamber. If there is an arterial pressure alarm during this procedure as a result of the access/arterial line being clamped the blood pump will stop, ensure to reset the alarm to allow the blood pump to run for enough time for the blood to be removed from the line and continue as the steps detailed above. Ensure to always attach the pressure line back on before unclamping the arterial/access line.

If the blood is already clotted in the pressure line DO NOT return it into the machine, leave it as it is as you do not want to return clots to the chamber. If blood reaches the pressure measurement sensor at the top of the pressure line you may experience an arterial pressure measurement failure alarm. Attempt to remove the blood if not clotted following the steps above, and reset the alarm. If the alarm does not reset the kit may require to be changed.



COMMON TROUBLESHOOTING DURING THERAPY

SOLUTION FLOW/PUMP FAILURE/FLOW PROBLEM

Ensure to diagnose and rectify as soon as possible to avoid a fluid balance management error. Often these alarms can be triggered by unbroken pins. Always ensure pins are broken, at times they can seem broken, however if they are not fully open the machine is unable to pull the fluid at the desired rate and therefore an alarm is triggered, which will then trigger subsequent alarms. Ensure to double check pins/clamps if any of these alarms are triggered to ensure this is not the problem.

OMNI CLOTTING

This can happen for various reasons, around 50% of these being access issues – please speak to medical staff regarding any access problems. The OMNI cannot run successfully if the access is poor. Several blood pump or therapy stops will cause the OMNI to clot. Ensure not to press the 'BLOOD PUMP' and 'THERAPY' buttons on and off when in therapy. Ensure anticoagulation is appropriate.

SELF TESTS

The machine carries out tests every 24 hours during therapy, it will alarm and flash, leave it to go through this process. If another alarm triggers during this time, it may cause the tests to fail. Press reset alarm and allow the tests to continue, this should resolve this issue. Please note that pressing the blood pump icon off can block the 24 hour self test.



COMMON TROUBLESHOOTING DURING THERAPY

SOLUTION CHAMBER REGULATION FAILURE

The most common reason for this alarm is due to a pin not broken properly on the dialysate bag or there is a clamp left on, causing a level regulation problem in the solution chamber. Check the pin and clamps and reset the alarm, it can take 2 or 3 alarm resets for the dialysate fluid level to resolve, giving it time to flow through the warmer bag to the solution chamber. During the problem resolution if the alarm does not reset after 2 or 3 times, it may be required to release the green solution pressure line for a second by disconnecting from the green pressure port at the side of the machine and connecting it securely back on to the green solution pressure port and then resetting the alarm.

Additionally, if not a pin or clamp it could be due to condensation within the solution chamber. If condensation is noted, go to the 'levels' page where you see the 4 chambers. Going to the solution chamber, keeping it in auto regulation, slowly increase the level in the chamber to get rid of the bubbles, the machine should bring the level back down automatically, clearing the bubbles and resolving the issue. Press reset alarm, and allow the machine to continue with therapy automatically.



TEMPORARY DISCONNECTION AND RECIRCULATION



This is for when the patient will be off the OMNI for a short period.



Only press temporarily disconnect patient when ready to return the blood, with equipment and sterile field prepared. Once this is pressed, the blood pump stops and the blood will begin to clot.



Return the blood as listed on the screen, ensuring to follow all steps in turn.



Put into recirculation once blood returned, connecting the blue and red lines to the 3 way tap.



Once ready to reconnect to the patient, ensure there are no clots within the set, if so a new set may be required as we cannot risk returning blood clots to the patient.



Follow the steps on screen and reconnect the patient.



If recirculation following temporary disconnection has been done for longer than the recommended recirculation time then we must change to a new set.



CHANGE OF KIT



This is used when we are taking the current kit off and a new one is required.



Follow the steps for returning the blood (ensuring to only press the Change Kit button when prepared to return the blood).



Once completed, the OMNI will then take you to setting up a new kit. Ensure to follow all steps in turn.



ENDING THERAPY



For ending the therapy return blood as stated following all steps in turn, reading the screen carefully and ensuring again to only press the end therapy button when ready to do so to avoid the blood clotting prior to blood return.



If ending therapy due to being unable to continue therapy (for example access issues), the machine will only return blood if it is able to do so, it does not return clotted blood.



Once the blood is returned ensure to take the RETURN/BLOOD line off the patient and make sure that the patient is fully disconnected before pressing unload disposables.

UNLOAD DISPOSABLES



Follow these steps on screen. Pressure lines do not get a tick until unloaded. Blood or fluid side door alarm on unload is when the doors open due to the force of the pumps unloading. Hold your finger on the door that is open and press reset alarm, unload will continue and complete. If there is any other alarm, ensure to reset this alarm to allow the kit to unload. It will not unload if there is an alarm to be dealt with.



Once unloaded click NEW THERAPY and this will take you back to the home page where you can switch the machine off or visit the HISTORY page.

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